Lab Report – EE-170L Computer Programming

NAME	Danyal Hassan
ID	750
SEMESTER	2nd
LAB	Programming
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COMPUTER PROGRAMMING

Electrical Engineering Department — UET Nowshera

LAB 6

Lab Title: The for Loop in C++

1. OBJECTIVE

The objectives of this lab are:

- To understand the syntax and working of the for loop in C++.
- To examine how a nested for loop functions and when it is appropriate to use one.
- To implement programs that use for loops for counting, pattern printing, and computing sums.
- To apply nested for loops to generate multiplication tables and triangular patterns.

2. THEORY

The for loop is a repetition control structure designed for situations where the number of iterations is known in advance. It combines initialization, condition checking, and variable update into a single compact line, making it well suited for counter-driven repetition.

Syntax:

```
for (initialization; condition; update)
{
    // statements to repeat
}
```

Flow of Execution:

- The initialization expression runs once at the start, setting up the loop counter.
- The condition is evaluated before each iteration. If true the body executes; if false the loop ends.
- After each iteration the update expression runs, usually incrementing or decrementing the counter.
- This cycle repeats until the condition becomes false.

Nested for Loop

A nested for loop is a for loop placed inside another for loop. For every single iteration of the outer loop, the inner loop completes all of its own iterations. This structure is commonly used for multiplication tables, two-dimensional data, and pattern printing.

Syntax concept:

```
for (int i = 0; i < rows; i++)       // outer loop
{
    for (int j = 0; j < cols; j++)   // inner loop
    {
        // inner loop statements
    } }
```

3. SOFTWARE USED

- DEV-C++ IDE
- C++ Compiler (bundled with DEV-C++)

4. TASKS

Task 1 — Display Numbers Using for Loop

A C++ program was written using a for loop to display the numbers 1 through 10, with each number printed on a separate line. The loop variable is initialized to 1, runs while it is 10 or less, and increments by one each iteration.

```
#include <iostream> using
namespace std;


int main()
{
    // loop from 1 to 10 and print each value
for (int i = 1; i <= 10; i++)
    {
        cout << i << endl;
    }
    return 0;
}
```

Task 2 — Multiplication Table Using Nested for Loop

A C++ program was written using nested for loops to generate a complete multiplication table from 1 to 10. The outer loop iterates over each row and the inner loop iterates over each column, printing the product formatted with tab spacing so the table aligns cleanly on screen.

```
#include <iostream>
```

```
using namespace std;

int main()
{
    // outer loop: rows 1 to 10
for (int i = 1; i <= 10; i++)
    {
        // inner loop: columns 1 to 10
for (int j = 1; j <= 10; j++)
        {
            cout << i * j << "\t"; // tab-separated product
        }
        cout << endl; // new line after each row
    }
    return 0;
}
```

Task 3 — Sum of Numbers Using for Loop

This program prompts the user to enter a positive integer and then uses a for loop to calculate the cumulative sum of all integers from 1 up to and including the entered value.

```
#include <iostream> using
namespace std;

int main()
{
    int num, sum = 0;     cout << "Enter a
positive integer: ";     cin >> num;

    // add each integer from 1 to num
for (int i = 1; i <= num; i++)
    {
        sum += i; // accumulate running total
    }
    cout  <<  "Sum  =  "  <<  sum  <<  endl;
return 0;
}
```

Task 4 — Triangle Pattern Using Nested for Loop

This program accepts the number of rows from the user and prints a right-angled triangle of asterisks. The outer loop controls the row number and the inner loop controls how many asterisks appear on each row, increasing by one with each successive row.

```
#include <iostream> using
namespace std;

int main()
{
    int rows;    cout << "Enter number
of rows: ";     cin >> rows;
```

```
    // outer loop: one pass per row
for (int i = 1; i <= rows; i++)
    {
        // inner loop: print i asterisks on this row
for (int j = 1; j <= i; j++)
        {
            cout << "* ";
        }
        cout << endl;
    }
    return 0;
}
```

5. CONCLUSION

This lab provided practical experience with the for loop and the nested for loop in C++. The for loop was applied to count-controlled repetition tasks including number sequences, cumulative sums, and user-determined pattern sizes. Nested for loops were used to produce a full multiplication table and a triangular asterisk pattern, demonstrating how two-dimensional iteration is structured. These constructs are a fundamental part of any C++ program that requires structured, controlled repetition.